PHISHING EMAIL ANALYSIS AND AWARENESS REPORT

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Course
B.E CSE (Cyber Security)

Internship
Cyber Security Internship – 2026

Organization
Future Interns

Date of Submission
April 2026

1. Executive Summary

This report presents an analysis of three email samples: one legitimate email, one confirmed phishing email, and one suspicious email containing a potentially malicious link.

The objective of this analysis is to identify phishing indicators, evaluate risks, and provide actionable recommendations to prevent similar attacks.

The findings show that phishing emails can bypass basic detection tools and rely heavily on social engineering techniques to deceive users.

2. Scope of Analysis

The analysis includes:

- Email header inspection
- Sender authentication validation (SPF, DKIM, DMARC)
- Domain and IP verification
- URL inspection using threat intelligence tools
- Identification of phishing indicators
- Risk classification

3. Tools & Technologies Used

Email Analysis Tools

- Google Message Header Analyzer
- MXToolbox Email Header Analyzer

Threat Intelligence Tools

- VirusTotal (URL scanning & reputation check)

Supporting Tools

- Web Browser Developer Tools
- Manual Header Inspection

4. Methodology

The following steps were followed during the analysis:

1. Collected email samples (legitimate and phishing)
2. Extracted and analyzed email headers
3. Verified authentication mechanisms (SPF, DKIM, DMARC)
4. Inspected sender domains and IP addresses
5. Analyzed embedded URLs using VirusTotal
6. Identified phishing indicators
7. Classified risk levels
8. Documented findings

5. Email Analysis Results

5.1 Sample 1 – Legitimate Email

Authentication Results:

- SPF: PASS
- DKIM: PASS
- DMARC: PASS

Observations:

- Sender domain matches infrastructure
- Trusted mail servers used
- No suspicious routing

Risk Level: SAFE

5.2 Sample 2 – Phishing Email (Header Failure)

Authentication Results:

- SPF: FAIL
- DKIM: FAIL
- DMARC: FAIL

Technical Findings:

- Sender domain spoofing detected
- Origin IP traced to unknown foreign server
- Return-Path mismatch observed
- Email originated from potentially compromised system

Risk Level: PHISHING

5.3 Sample 3 – Suspicious Email with URL

Suspicious URL Identified:

https://secure-bank-login-check.com/verify/account-update

VirusTotal Analysis:

- No immediate detection by antivirus engines
- Domain not widely flagged

Critical Observation:

- Domain impersonates a banking service
- Not an official or verified domain
- Combined with header failures → strong phishing indicator

Authentication Results:

- SPF: FAIL
- DKIM: FAIL
- DMARC: FAIL

Risk Level: SUSPICIOUS / HIGH RISK

6. Key Phishing Indicators Identified

- Domain spoofing (fake sender domain)
- Authentication failures (SPF, DKIM, DMARC)
- Suspicious or newly created domains
- Mismatch between sender and return-path
- Use of urgency (e.g., “Urgent”, “Immediate Action Required”)
- Presence of deceptive URLs
- Email routing through unknown servers

7. URL & Domain Analysis

URL Tested:

https://secure-bank-login-check.com/verify/account-update

Findings:

- Domain is not officially associated with any bank
- Uses deceptive naming pattern
- May evade detection in early stages
- Potential phishing landing page

Important Insight:
Even if VirusTotal shows a URL as “clean”, it does NOT guarantee safety.
New phishing domains often bypass detection initially.

8. Attack Lifecycle (Phishing Flow)

[Attacker]

↓

Creates Spoofed Email

↓

Adds Malicious Link

↓

Victim Receives Email

↓

Clicks Link

↓

Fake Website Opens

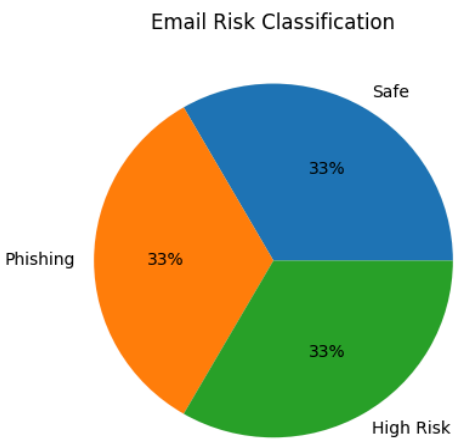
↓

Credentials Stolen

↓

Account Compromise

9. Risk Classification Summary



10. Recommendations & Prevention

Recommended Actions

- Always verify sender email domain
- Check authentication results (SPF, DKIM, DMARC)
- Hover over links before clicking
- Access services only through official websites
- Use email filtering and security solutions

Avoid

- Clicking unknown or suspicious links
- Sharing passwords or OTPs via email
- Trusting urgent messages without verification
- Downloading attachments from unknown senders

11. Appendix

A. Legitimate Email Header Analysis

MessageId	0987654321@mail.company.org
Created at:	4/16/2026, 9:15:07 PM GMT+5:30 (Delivered after 3 sec)
From:	"Company Alerts" <alert@company.org>
To:	userdemo@gmail.com
Subject:	Security Notification – Account Activity
SPF:	**pass** with IP 198.51.100.10 Learn more
DKIM:	**pass** with domain company.org Learn more
DMARC:	**pass** Learn more

#	Delay	From *			To *	Protocol	Time received
0	1 sec	smtp.company.org	→		mail.company.org	ESMTP	4/16/2026, 9:15:08 PM GMT+5:30
1	2 sec	mail.company.org.	→	[Google]	mx.google.com	ESMTPS	4/16/2026, 9:15:10 PM GMT+5:30
2			→	[Google]	2002:a17:906:abcd:b0:2f3:efgh:5678	SMTP	4/16/2026, 9:15:10 PM GMT+5:30

B. Phishing Email Header Analysis

MessageId	AABBCC1122@mail.secure-update.com
Created at:	4/22/2026, 9:44:28 PM GMT+5:30 (Delivered after 5 sec)
From:	"Secure Bank Alerts" <alerts@secure-bank.com>
To:	victimuser@gmail.com
Subject:	Action Required: Suspicious Activity Detected
SPF:	**fail** with IP Unknown! Learn more
DKIM:	**fail** with domain Unknown! Learn more
DMARC:	**fail** Learn more

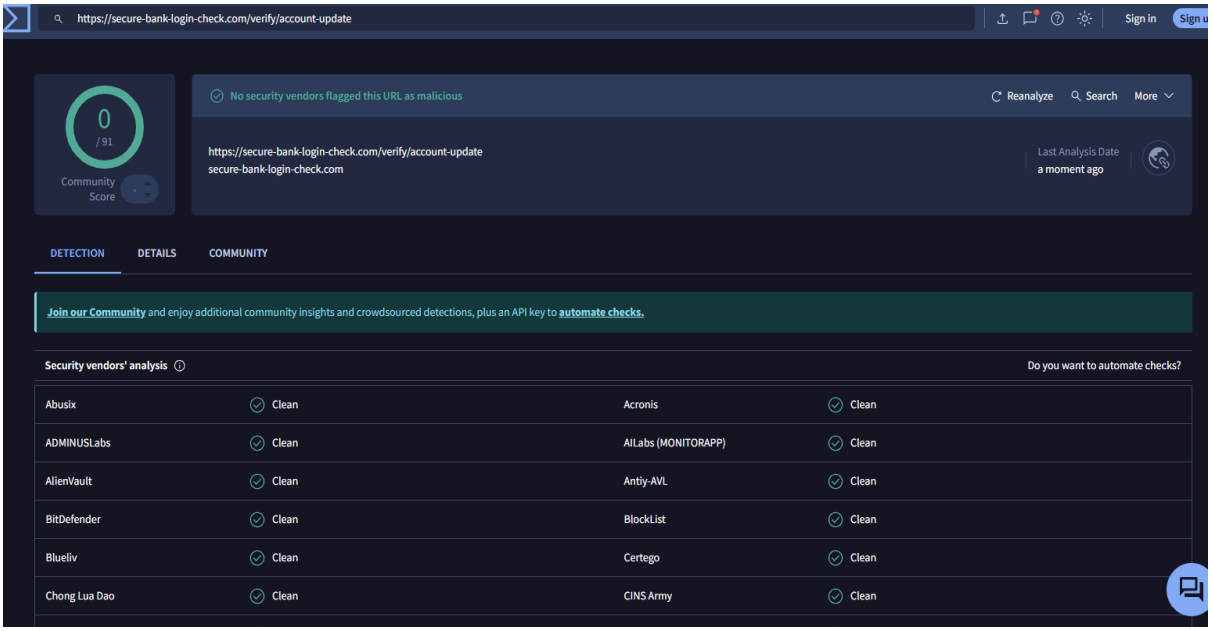
#	Delay	From *			To *	Protocol	Time received
0	1 sec	unknown	→		mail.secure-update.com	ESMTP	4/22/2026, 9:44:29 PM GMT+5:30
1	4 sec	mail.secure-update.com.	→	[Google]	mx.google.com	ESMTPS	4/22/2026, 9:44:33 PM GMT+5:30
2			→	[Google]	2002:a05:620a:1456:b0:123:abcd:5678	SMTP	4/22/2026, 9:44:33 PM GMT+5:30

C.Suspicious Email Header Analysis

MessageId	6677889900@unknown-server.ru
Created at:	4/21/2026, 11:35:08 PM GMT+5:30 (Delivered after 4 sec)
From:	"Bank Support" <support@secure-bank.com>
To:	victimuser@gmail.com
Subject:	URGENT: Verify Your Bank Account Immediately
SPF:	fail with IP Unknown! Learn more
DKIM:	fail with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From *			To *	Protocol	Time received
0	1 sec	unknown	→		unknown-server.ru	ESMTP	4/21/2026, 11:35:09 PM GMT+5:30
1	3 sec	unknown-server.ru.	→	[Google]	mx.google.com	ESMTPS	4/21/2026, 11:35:12 PM GMT+5:30
2			→	[Google]	2002:a05:aa1b:9999:b0:7b8:fake:1111	SMTP	4/21/2026, 11:35:12 PM GMT+5:30

D.URL Scan Result (VirusTotal)



VX Vault	Clean	Webroot	Clean
Xcitium Verdict Cloud	Clean	Yandex Safebrowsing	Clean
ZeroCERT	Clean	0xSI_f33d	Unrated
alphaMountain.ai	Unrated	AlphaSOC	Unrated
ArcSight Threat Intelligence	Unrated	AutoShun	Unrated
Bfore.Ai PreCrime	Unrated	Bkav	Unrated
ChainPatrol	Unrated	Cluster25	Unrated
Criminal IP	Unrated	CSIS Security Group	Unrated
Cyan	Unrated	Ermes	Unrated
Forcepoint ThreatSeeker	Unrated	GCP Abuse Intelligence	Unrated
GreyNoise	Unrated	Gridinsoft	Unrated
Guardpot	Unrated	Hunt.io Intelligence	Unrated
Kaspersky	Unrated	Lumu	Unrated
MalwareURL	Unrated	Mimecast	Unrated
Netcraft	Unrated	PhishFort	Unrated

12. Conclusion

This analysis demonstrates that phishing attacks rely heavily on deception rather than technical complexity.

Even when security tools do not immediately flag a threat, deeper analysis of email headers and domain behavior can reveal malicious intent.User awareness and proper verification practices remain the strongest defense against phishing attacks.